

Money Plot Labs Technical Whitepaper 002

The Asymmetric Glide Path: Solving Bifurcated Growth

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1 The Transcendental Challenge

In Technical Whitepaper 001, we derived a closed-form solution for the working years w under a unified growth rate r . However, modern portfolio theory suggests a "glide path," where risk—and therefore expected return—is reduced as an individual moves from accumulation to decumulation.

When we introduce two distinct rates, r_{acc} (Working Growth) and r_{dec} (Retirement Growth), the boundary equation becomes:

$$R_f(A_w(w, r_{acc}), f, r_{dec}) = L \quad (1)$$

Expanding this using the Annuity Due model derived previously:

$$\left[A_0(1 + r_{acc})^w + s \frac{(1 + r_{acc})^w - 1}{r_{acc}} \right] (1 + r_{dec})^{H-w} - c \frac{(1 + r_{dec})^{H-w+1} - (1 + r_{dec})}{r_{dec}} = L \quad (2)$$

Unlike the unified case, the variable w now appears in the exponents of two different bases: $(1 + r_{acc})$ and $(1 + r_{dec})$. This is a **transcendental equation**. There is no finite sequence of algebraic operations that can isolate w , necessitating a shift from analytical solvers to numerical approximation.

2 The Numerical Solver: Bisection Method

To solve for w , the Money Plot engine utilizes the Bisection Method (a form of Binary Search). This method is robust, guaranteed to converge on a continuous function, and avoids the "division by zero" or "log of negative" errors that Newton-Raphson methods might encounter in extreme financial scenarios.

2.1 The Objective Function

We define the function $f(w)$ as the net difference between the terminal balance and the target legacy floor:

$$f(w) = \text{TerminalBalance}(w) - L \quad (3)$$

The goal is to find the root where $f(w) = 0$ within the interval $[0, H]$.

2.2 The Algorithm

1. **Initialize:** Set $low = 0$, $high = H$.
2. **Evaluate Boundaries:**
 - If $f(0) \geq 0$, the individual is "Already Free" ($w = 0$).
 - If $f(H) < 0$, the target is unreachable ("Never").
3. **Iterate:** For $i = 1$ to 60:
 - $mid = \frac{low+high}{2}$
 - If $f(mid) < 0$, then $low = mid$ (Must work longer).
 - Else, $high = mid$ (Can retire sooner).
4. **Result:** $w = \frac{low+high}{2}$.

A 60-iteration search provides a precision of $H \cdot 2^{-60}$. Over a 100-year horizon, this yields a precision of approximately 8.6×10^{-17} years, far exceeding any practical requirement.

3 Bifurcated Model Components

3.1 Accumulation (The Input)

The asset state at the moment of retirement (A_w) is governed by the accumulation rate:

$$A_w = A_0(1 + r_{acc})^w + s \frac{(1 + r_{acc})^w - 1}{r_{acc}} \quad (4)$$

3.2 Decumulation (The Output)

The depletion of A_w over the retirement years $f = H - w$ is governed by the decumulation rate using the Annuity Due timing (spending at the start of the year):

$$Balance_t = A_w(1 + r_{dec})^t - c \frac{(1 + r_{dec})^{t+1} - (1 + r_{dec})}{r_{dec}} \quad (5)$$

4 Impact of Rate Bifurcation

Splitting the rates reveals two critical insights:

1. **Accumulation Dominance:** Higher r_{acc} reduces w exponentially. A 1% increase in r_{acc} during early working years has a larger impact on the retirement date than a 1% increase in r_{dec} during retirement.
2. **Sequence Protection:** Lowering r_{dec} (e.g., shifting to bonds) increases the required A_w but flattens the depletion curve, reducing the variance of the terminal outcome L .